KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer ANY THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer ANY THREE questions from this section in Script A)

1. a) Define the terms: 'confusion', 'diffusion', 'cryptography' and 'cryptanalysis'. Discuss Shannon characteristics of good cipher. (12)
   b) 'RSA is a commutative like cryptosystem' – how? Explain deliberately. (08)
   c) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   d) Define 'known-plaintext' attack. How this attack can mount over RSA? (07)

2. a) Discuss characteristics of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Digital signature and public key cryptography can be combined – why and how? Explain. (08)
   c) Using an example explain ElGamal cryptosystem. How does it relate with discrete logarithm problem? (10)
   d) Discuss threats to e-mail. (05)

3. a) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   b) Explain AES algorithm using a block diagram. Make a comparison between DES and AES. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (09)

4. a) What is meant by homomorphic encryption? Show that RSA and ElGamal cryptosystems are homomorphic. (10)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) Explain the mechanism of triple DES algorithm. (05)

SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) What is salt and how does it make password hashing more secure? Also explain rainbow table attack and how salt can help against rainbow table attack. (15)
   b) How does the attacker know what algorithm and salt to use in a dictionary attack? (06)
   c) Explain Buffer Overflow attack with proper code and stack diagram. Also discuss about the countermeasures that are used against Buffer Overflow attack. (14)

6. a) Explain how Format String attack works with appropriate activation record diagrams. Also explain different uses of Format String attack with proper example(s) and stack diagrams. (15)
   b) Demonstrate Integer Overflow attack using suitable example(s). (07)
   c) Explain SQL Injection attack with practical example. Also discuss about the prevention techniques used against SQL Injection attack. (13)

7. a) Explain Persistent, Reflected and DOM-based Cross-Site Scripting (XSS) attacks with example(s). (15)
   b) Explain the methods of preventing XSS attacks. (10)
   c) What are web bugs? Explain how attackers bypass CAPTCHA protection. (10)

8. a) Explain how combining Paging with Segmentation provides better performance for protection in general purpose operating system compared to using them separately. (09)
   b) What is a Covert Channel? How to create Covert Channels? (06)
   c) Briefly explain the following attacks: (i) Ransomware, (ii) Salami Attack, (iii) Tapjacking, (iv) Document virus using Microsoft's Dynamic Data Exchange. (16)
   d) What is Access Control Directory? (04)